Posting Like a State: How do Citizens Interact with Autocrats' Media

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Abstract

How should scholars interpret engagement with official media in authoritarian settings? Existing accounts often read digital traces on state platforms either as evidence of propaganda success or as noise generated by skeptical audiences. I argue instead that official accounts are mixed channels that bundle public service information with political messaging, and that citizens sort their responses according to the visibility of the expressive act. Using 4.21 million post titles from 329 Chinese local-government WeChat accounts, I first show that public service content dominates citizen attention, creating an attention dilemma: service information helps governments build durable audiences for future political communication while crowding out overt propaganda at the moment of consumption. I then exploit two interface reforms that changed the meaning of a click. The 2018 redesign collapsed low-visibility likes and rebundled approval with socially legible recommendation, sharply reallocating reactions across channels. The 2020 return of the like button provides cleaner causal leverage: like rates jump for hard and soft propaganda, while public-signal rates remain flat. These findings recast engagement metrics as strategic behavioral traces rather than direct measures of regime support and extend preference falsification theory to layered digital expression.

Keywords: digital authoritarianism; propaganda; official social media; WeChat; preference falsification; state–society interaction.

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1 Introduction

How do citizens in autocracies interact with official social media, and what do those traces actually mean? As propaganda, public communication, and service delivery migrate onto digital platforms, state-controlled accounts accumulate vast numbers of reads, likes, and shares. These metrics are tempting to interpret as indicators of propaganda success or regime support. Yet in authoritarian settings, interacting with official content is rarely a purely expressive act. Citizens may engage because they need information, because a post is difficult to avoid, because approval is low-cost, or because visible endorsement carries reputational and political risk. The political meaning of digital engagement is therefore fundamentally ambiguous.

Existing work speaks to pieces of this puzzle, but rarely to the mechanism at stake here. Research on Chinese propaganda shows that the party-state competes for attention through commercialization, soft propaganda, distraction, and popularization rather than relying on crude indoctrination alone (Brady 2008; King, Pan, and Roberts 2017; Lu and Pan 2021; Mattingly and Yao 2022; Repnikova and Fang 2018; Roberts 2018; Stockmann 2013; Wang 2024; Zou 2021). Research on preference falsification and self-censorship emphasizes the gap between private belief and public expression under pressure (Carter, Carter, and Schick 2024; Jiang and Yang 2016; Kuran 1995; Noelle-Neumann 1993; Shen and Truex 2020). A third literature shows that government WeChat accounts are not merely propaganda outlets: local authorities also use them for public service delivery, routine contact, and audience acquisition (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026; Zheng 2013; Zheng and Zheng 2014). A related strand highlights semi-official accounts, delegated censorship, and layered platform governance, suggesting that state influence is often embedded in organizationally ambiguous or platform-native forms (Neagli 2021; Sun and Zhao 2022; Tai and Fu 2020). What these literatures do not yet explain well is how platform design changes the political meaning of interaction itself.

I argue that official social media in digital authoritarianism are mixed channels that bundle public service information with political messaging, and that citizens sort their behavior across different levels of expressive risk. On WeChat, reading is low-risk consumption, a like is low-

visibility approval, *zaikan* is socially legible endorsement, and sharing is the highest-visibility form of amplification. This layered-expression perspective shifts the theoretical question away from whether citizens "believe" propaganda and toward how they manage observable traces of alignment.

I evaluate this argument using 4.21 million post titles published by 329 prefecture-level government WeChat official accounts between 2015 and 2024. The article makes two empirical moves. First, I use a theory-driven 'LLM + human' classification workflow to assign fine-grained post labels and then collapse them into three main content families: hard propaganda, soft propaganda, and public service. Second, I exploit the 2018 and 2020 WeChat interface reforms as discontinuities that changed what a click meant. The 2018 redesign bundled approval with public recommendation; the 2020 redesign restored a lower-visibility like button alongside *zaikan*.

Three findings organize the paper. Public service content dominates attention, creating an attention dilemma: the content that helps governments build durable audiences is not the content that looks most overtly political. The 2018 redesign sharply contracts low-visibility likes and reallocates reactions into a more socially legible channel. The 2020 return of likes provides cleaner causal leverage: like rates jump for hard and soft propaganda, while public-signal rates remain flat. Official engagement metrics are therefore better understood as strategic behavioral traces than as transparent measures of regime support.

This article makes three contributions. First, it extends preference falsification theory from speech to platform-mediated behavior by showing that digital expression is layered across reading, liking, *zaikan*, and sharing. Second, it contributes to the study of propaganda and media consumption by separating attention from endorsement and by showing how service content both builds and competes for political attention. Third, it recasts official social media as an everyday site of authoritarian state–society interaction, where governments manage not only information but also the terms under which citizens can safely engage.

2 Theory and Hypotheses

Official social media in authoritarian settings should be understood as a platformed interaction regime rather than as a simple channel of state persuasion. In classic preference falsification accounts, the central distinction is between private preference and public statement. On platforms such as WeChat, however, expression is not binary. The interface decomposes response into multiple actions that expose users to different audiences and therefore to different expressive risks. The state may observe aggregate traces, but friends, colleagues, and acquaintances can also infer meaning from visible acts. Digital preference falsification is therefore layered: the same user may privately consume official content, register low-visibility approval, and avoid socially visible endorsement (Kuran 1995; Noelle-Neumann 1993; Shen and Truex 2020; Tsai and Men 2018; Xu 2022).

This layered-expression logic matters because Chinese government WeChat accounts are mixed channels rather than purely political outlets. Existing research shows that state agencies use social media not only to disseminate political messages but also to provide services, route transactions, manage crises, and sustain routine interaction with the public (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026; Zheng 2013; Zheng and Zheng 2014). That literature is especially useful because it shows that many users approach official accounts as lookup tools or service interfaces rather than as ideological venues. Service content therefore helps governments build durable audiences, normalize repeated contact, and increase the probability that users encounter future political content. At the moment of consumption, however, service information also competes with overt political messaging for scarce attention. This is the paper's attention dilemma.

The same logic also changes how we should think about propaganda. Contemporary influence online often works through participatory persuasion, semi-official intermediaries, and delegated forms of control rather than through a single visible propaganda apparatus (King, Pan, and Roberts 2017; Lu and Pan 2021; Neagli 2021; Repnikova and Fang 2018; Sun and Zhao 2022). Hard propaganda foregrounds hierarchy, ideology, discipline, and coercive state capacity. Soft propaganda packages regime-favorable messages in more appealing, emotionally resonant, or developmental

frames (Mattingly and Yao 2022; Pan, Shao, and Xu 2022; Wang 2024). Citizens may read all three kinds of content, but the willingness to leave an observable trace should vary with both content risk and interface design. Platform reforms do not merely change aesthetics. They reprice the political and social costs of interacting with the state.

This framework yields four expectations:

H1 (Attention and Utility). Public service content should attract more routine consumption than either hard or soft propaganda because it provides immediate utility and lower expressive stakes.

H2 (Rebundling Visible Approval). When the 2018 redesign bundles approval with public recommendation, low-visibility like responses should collapse and observed reactions should be reallocated into more socially legible channels.

H3 (Private Substitution). When low-visibility likes return in 2020, approval should shift back toward likes rather than rising uniformly across all public-facing reactions.

H4 (Political-Risk Gradient). These reallocations should be strongest for hard propaganda, then soft propaganda, and weakest for routine public-service content.

Table 1: Platform reforms and theoretical expectations

Time point	Brief policy content	Reading / exposure	Visible endorsement	Low-risk approval
2017–05 (v6.5.8 rollout)	WeChat introduces "See" (*kanyikan*), adding a friend- and algorithm-based discovery channel while leaving the bottom "like" button unchanged.	+	0	0
2018–12–21	WeChat 7.0 replaces "like" with "good-looking" (*haokan*, later *zaikan*), bundling approval with recommendation into "See" (*kanyikan*).	0	reallocated / ambiguous	–
2020–06–29	The "like" button is restored at the bottom of public-account articles alongside the existing "seen" (*zaikan*) and share buttons.	0	0/–	+

Note: '+' indicates an expected increase, '-' an expected decrease, and '0' little immediate change. The 2018 redesign mechanically rebundles approval with recommendation, so observed public-signal measures may be reallocated rather than move with a single clean sign. The strongest directional effects are expected for hard propaganda.

3 Setting, Data, and Measurement

WeChat is the central digital infrastructure of everyday life in China. Since its launch in 2011, it has evolved into a multipurpose "super app" that combines messaging, payments, news distribution, and social networking within a single platform. By the 2020s, it had become the primary interface through which Chinese citizens communicate, consume news, and access digital public and private services. For a study of official digital communication in China, WeChat matters not because it is just another social media platform, but because it is deeply embedded in ordinary social, administrative, and economic life (Ng 2015; Roberts 2020).

Within this ecosystem, "public accounts" operate as mixed institutional channels. They transmit propaganda and political messaging, publicize leadership priorities, and provide highly practical information and service announcements (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026; Zheng 2013; Zheng and Zheng 2014). At the same time, the broader WeChat ecosystem includes semi-official and organizationally ambiguous accounts that can carry state-aligned messages without looking fully official (Neagli 2021). A typical article displays a read counter together with several engagement mechanisms: a like button, *haokan*/*zaikan*, sharing, collecting, and comments. These buttons do not measure the same thing with different labels. They ask users to leave different kinds of traces under different conditions of visibility.

The key institutional changes are the 2018 and 2020 interface reforms. In December 2018, WeChat replaced the original like button on newly pushed public-account articles with a "good-looking" button tightly integrated with the "See" tab. A previously quasi-private like was transformed into a socially legible recommendation. In June 2020, WeChat restored the traditional like button alongside *zaikan* and sharing, thereby unbundling low-visibility approval from public recommendation. I treat 2017 as background discovery reform and 2018/2020 as the two main discontinuities for expressive engagement. Table 1 summarizes these shifts.

This study uses the universe of posts published by 329 prefecture-level government WeChat public accounts between 1 January 2015 and 31 December 2024, covering 31 provincial-level units and 4,207,072 posts. The unit of observation for classification is the individual post title. For

each post, I record the account identifier, publication date, title, and engagement indicators displayed in the interface, including reads, likes, shares, comments, and *haokan*/*zaikan*. The archived source file also preserves a first-pass category field, a short reason field, a keyword field, and a confidence score from an earlier coarse classification pass; these metadata are used only for title-based normalization and coding, not as outcomes. Because WeChat top-codes some counters at 100,000 (displayed as "100k+"), I recode those values to 100,001 so they remain ordered while preserving censoring at the upper bound.

To categorize post content at scale, I use an LLM as a text coder rather than as a generator. The production prompt is frozen before full-corpus assignment and uses only title-side inputs: title, account name, province, city, publication date, the archived raw category, keywords, and the short reason field. Engagement outcomes, account-level outcome patterns, and reform-period indicators never enter the classification prompt. The coding workflow is built on a fixed ten-category taxonomy and a transparent normalization layer for raw labels. The main text relies on this workflow only for measurement credibility; the SI documents taxonomy construction, raw-label normalization, coder-audit rules, and validation diagnostics.

I validate the LLM labels against a dual-coded gold-standard sample of 2,000 post titles coded under the same codebook and adjudicated to a final reference label. The SI reports the reliability protocol, aggregate validation metrics, class-wise performance, and a family-level confusion matrix. For the main analyses, I collapse the fine-grained labels into three theoretically motivated families: hard propaganda, soft propaganda, and public service. Hard propaganda includes overtly political and disciplinary content; soft propaganda packages regime-favorable messages through development, culture, tourism, and performance narratives; public service covers instrumental governance content such as procedures, welfare, health, traffic, weather, and emergency guidance. State-governance categories remain in the data for balance checks and appendix material but are not the paper's main theoretical families.

The paper's measurement logic is therefore theory-driven rather than purely unsupervised. The LLM solves the assignment problem at the post level; it does not decide which families matter

theoretically. That distinction is important because the theoretical categories come from prior literature on mixed governance communication, soft propaganda, and expressive visibility rather than from an inductive topic model alone.

4 Descriptive Findings

The descriptive evidence establishes three facts that motivate the causal analysis. First, official WeChat feeds are genuinely mixed channels. Across the full corpus, public service is the largest of the three main families with about 1.34 million posts, followed by soft propaganda with 1.13 million posts and hard propaganda with 1.00 million posts. Governments are not using these accounts only to broadcast ideology; they are also using them to solve everyday informational problems.

Second, attention is even more skewed toward service content than production is. Public service posts receive about 432 average reads, compared with roughly 218 for soft propaganda and 135 for hard propaganda. Public service therefore draws more than three times the average readership of hard propaganda. This is the paper's attention dilemma: the content that makes official accounts useful and sticky is also the content that crowds out overtly political messaging at the moment of consumption.

Third, reaction profiles shift sharply around interface changes. Like rates collapse after the 2018 redesign and public-signal rates rise across all three main families, indicating that a once-private trace has been rebundled into a more socially legible recommendation channel. When likes return in 2020, like rates recover while public-signal rates do not exhibit corresponding jumps. The descriptive patterns therefore already suggest that the interface changes the form of engagement more than the underlying presence of attention.

5 Research Design

The main empirical strategy is a local linear regression discontinuity design centered on the two interface reforms that directly changed the meaning of approval: 21 December 2018 and 1 July 2020. I aggregate the data to day-by-content-group cells and estimate discontinuities in five outcomes: reads per post, like rates, *zaikan* rates, public-signal rates (shares plus *zaikan*), and total reaction rates. The running variable is days from the cutoff. All specifications use local linear fits with post-count weights and a one-year window around the reform.

I also report two simple robustness checks. First, I re-estimate the main specifications using a narrower 180-day bandwidth and a seven-day donut around the cutoff. Second, I examine continuity in post volume and each content group's daily share of total posting. These checks matter because the 2018 redesign changes the observable meaning of approval itself, while the 2020 redesign more cleanly restores an additional low-visibility channel.

This difference in institutional interpretation matters for how I read the estimates. The 2018 discontinuity is informative about rebundling and expressive reallocation, but it is not a clean test of net endorsement because the button semantics change. The 2020 discontinuity is the cleaner causal test of private substitution: it restores a low-visibility channel while leaving higher-visibility channels in place. The strongest causal claims therefore come from whether approval moves back into likes without a parallel jump in public-signal outcomes.

6 Causal Findings

The 2018 redesign behaves like a rebundling shock. Across hard propaganda, soft propaganda, and service content, low-visibility like rates fall sharply at the cutoff. The robust local-linear estimates are about -0.0098 for hard propaganda, -0.0129 for soft propaganda, and -0.0079 for service. At the same time, *zaikan* and public-signal rates jump: for hard propaganda the public-signal estimate is about $+0.0068$, for soft propaganda about $+0.0068$, and for service about $+0.0037$. Read counts per post show no clear discontinuity. Substantively, once approval is bundled with

public recommendation, observable reactions are reallocated into a more socially legible channel rather than disappearing altogether.

The 2020 redesign is cleaner and more directly speaks to layered expression. When WeChat restores the like button, like rates jump for all three main families, but the increase is largest for hard propaganda (+0.0046), followed by soft propaganda (+0.0039), and then service (+0.0031). By contrast, public-signal rates around the same cutoff are statistically flat for hard propaganda, soft propaganda, and service. For hard and soft propaganda, total reaction rates also rise, while service shows a smaller and less precise change. This is the paper's main causal result: when a lower-visibility approval channel returns, citizens use it, especially for political content, without producing an equivalent increase in more public-facing endorsement.

The 2020 results also help separate attention from endorsement. Reads per post do not increase for hard or soft propaganda and actually decline for service content at the cutoff. Whatever else is changing in 2020, the core discontinuity is not a sudden demand shock for official information. It is a reallocation of approval into a lower-risk expressive form.

7 Heterogeneity

The clearest heterogeneity is by content risk. The 2020 like-rate rebound follows the expected ordering from the theory: hard propaganda reacts most strongly, soft propaganda next, and service least. This ranking is consistent with the claim that the political meaning of an observable trace matters most when the underlying content is overtly political.

At the same time, the continuity checks caution against treating every discontinuity symmetrically. Around the 2018 redesign, hard-propaganda post volume and posting share exhibit some discontinuity, which makes the 2018 estimates best read as evidence of expressive rebundling rather than a standalone clean causal effect. Around the 2020 redesign, continuity is considerably cleaner for hard propaganda and service, while soft-propaganda posting share shifts more noticeably. The strongest identification therefore comes from the comparison between hard propaganda

and service at the 2020 cutoff.

Taken together, these heterogeneities sharpen the substantive interpretation. Platform reforms do not affect all state content equally. They matter most when citizens are asked to leave politically interpretable traces and when the interface offers meaningful choice between safer and riskier forms of assent.

8 Conclusion

Official engagement metrics on authoritarian platforms are not transparent measures of support. They are shaped by mixed content environments and by the visibility structure of the interface itself. On WeChat, public service content sustains attention and builds durable audiences, while interface design determines how safely citizens can convert attention into approval.

The broader implication is that digital authoritarian control works partly by structuring traces, not only by changing beliefs. Once we distinguish low-risk consumption from low-visibility approval and from socially legible endorsement, the political meaning of a click changes. That insight extends preference falsification theory to platform-mediated behavior and suggests a more cautious reading of engagement metrics in authoritarian settings.

References

Brady, Anne-Marie. 2008. *Marketing Dictatorship: Propaganda and Thought Work in Contemporary China.* Lanham, MD: Rowman & Littlefield.

Carter, Erin Baggott, Brett L. Carter, and Stephen Schick. 2024. "Do Chinese Citizens Conceal Opposition to the CCP in Surveys? Evidence from Two Experiments." *The China Quarterly* 259:804–813.

Jiang, Junyan, and Dali L. Yang. 2016. "Lying or Believing? Measuring Preference Falsification From a Political Purge in China." *Comparative Political Studies* 49 (5): 600–634.

Jiang, Tingting, Ying Wang, Tianqianjin Lin, and Lina Shangguan. 2021. "Evaluating Chinese Government WeChat Official Accounts in Public Service Delivery: A User-Centered Approach." *Government Information Quarterly* 38 (1): 101548.

King, Gary, Jennifer Pan, and Margaret E. Roberts. 2017. "How the Chinese Government Fabricates Social Media Posts for Strategic Distraction, Not Engaged Argument." *American Political Science Review* 111 (3): 484–501.

Kuran, Timur. 1995. *Private Truths, Public Lies: The Social Consequences of Preference Falsification.* Cambridge, MA: Harvard University Press.

Liu, Tzu-Hsuan, Zhihao Ma, and Yiwei Xia. 2024. "Serving on WeChat: Understanding the Logics of Police's Engagement with the Public in Chinese Contexts." *International Journal of Law, Crime and Justice* 77:100665.

Lu, Yingdan, and Jennifer Pan. 2021. "Capturing Clicks: How the Chinese Government Uses Clickbait to Compete for Visibility." *Political Communication* 38 (1–2): 23–54.

Mattingly, Daniel C., and Elaine Yao. 2022. "How Soft Propaganda Persuades." *Comparative Political Studies* 55 (9): 1569–1594.

Neagli, Jackson Paul. 2021. "Grassroots, Astroturf, or Something in Between? Semi-Official WeChat Accounts as Covert Vectors of Party-State Influence in Contemporary China." *Journal of Current Chinese Affairs* 50 (3): 305–332.

Ng, Jason Q. 2015. *Politics, Rumors, and Ambiguity: Tracking Censorship on WeChat's Public Accounts Platform.* Research Report No. 57. Accessed March 26, 2026. The Citizen Lab, University of Toronto. https://citizenlab.ca/research/tracking-censorship-on-wechat-public-accounts-platform/.

Noelle-Neumann, Elisabeth. 1993. *The Spiral of Silence: Public Opinion—Our Social Skin.* 2nd ed. Chicago: University of Chicago Press.

Pan, Jennifer, Zijie Shao, and Yiqing Xu. 2022. "How Government-Controlled Media Shifts Policy Attitudes Through Framing." *Political Science Research and Methods* 10 (2): 317–332.

Repnikova, Maria, and Kecheng Fang. 2018. "Authoritarian Participatory Persuasion 2.0: Netizens as Thought Work Collaborators in China." *Journal of Contemporary China* 27 (113): 763–779.

Roberts, Margaret E. 2018. *Censored: Distraction and Diversion Inside China's Great Firewall.* Princeton, NJ: Princeton University Press.

Roberts, Margaret E. 2020. "Resilience to Online Censorship." *Annual Review of Political Science* 23 (1): 401–419.

Shen, Xiaoxiao, and Rory Truex. 2020. "In Search of Self-Censorship." *British Journal of Political Science* 51 (4): 1672–1684.

Stockmann, Daniela. 2013. *Media Commercialization and Authoritarian Rule in China.* Cambridge: Cambridge University Press.

Sun, Taiyi, and Quansheng Zhao. 2022. "Delegated Censorship: The Dynamic, Layered, and Multistage Information Control Regime in China." *Politics & Society* 50 (4): 583–607.

Tai, Yun, and King-wa Fu. 2020. "Specificity, Conflict, and Focal Point: A Systematic Investigation into Social Media Censorship in China." *Journal of Communication* 70 (6): 842–867.

Tsai, Wan-Hsiu Sunny, and Linjuan Rita Men. 2018. "Social Messengers as the New Frontier of Organization-Public Engagement: A WeChat Study." *Public Relations Review* 44 (3): 419–429.

Wang, Clyde Yicheng. 2024. "The Ideology Is Blowing in the Wind: Managing Orthodoxy and Popularity in China's Propaganda." *Political Communication* 41 (3): 435–460.

Xu, Vivien Weiwei. 2022. "WeChat and the Sharing of News in Networked China." *Digital Journalism* 10 (9): 1441–1463.

Yu, Jianjun, and Shuyuan Shen. 2026. "Serve the People: Public Service Information and Communication Strategies of Local Governments in China." Published online March 6, 2026, *Journal of Contemporary China,* 1–20.

Zheng, Lei. 2013. "Social Media in Chinese Government: Drivers, Challenges and Capabilities." *Government Information Quarterly* 30 (4): 369–376.

Zheng, Lei, and Tuo Zheng. 2014. "Innovation Through Social Media in the Public Sector: Information and Interactions." *Government Information Quarterly* 31 (S1): S106–S117.

Zou, Sheng. 2021. "Restyling Propaganda: Popularized Party Press and the Making of Soft Propaganda in China." *Information, Communication & Society* 26 (1): 201–217.

Online Supplemental Information

A WeChat Title Data and Classification Validation **SI-1**

A.1 Corpus and Coding Inputs SI-1

A.2 Taxonomy Construction and Discovery Review SI-1

A.3 Human Coding Rules and Validation Sample SI-3

A.4 Frozen LLM Prompt SI-3

A.5 Validation Results SI-4

Supplemental Information Roadmap. This SI section documents the construction and validation of the WeChat title-classification workflow used in the paper. It proceeds from corpus coverage and coding inputs to taxonomy construction, coder rules, the frozen production prompt, and validation against the adjudicated audit sample.

A WeChat Title Data and Classification Validation

A.1 Corpus and Coding Inputs

This section documents the title-based WeChat classification pipeline used in the main analyses. The cleaned corpus covers all posts published by 329 prefecture-level government WeChat public accounts between 1 January 2015 and 31 December 2024. Table SI.1 summarizes corpus coverage and the text-side inputs available for classification.

Table SI.1: Corpus coverage and coding inputs for the WeChat title dataset.

metric	value
Posts in cleaned corpus	4,207,072
Public accounts	329
Cities	329
Provincial-level units	31
Date range	2015-01-01 to 2024-12-31
Distinct raw category strings	862
Final detailed labels	10
Final higher-order families	4
Primary text field used for coding	Post title

The key measurement constraint is that the archived source file preserves post titles consistently across the full nationwide collection but does not preserve article bodies in the current working archive. The unit of text classification is therefore the individual post title. In addition to the title itself, the archive preserves several title-side metadata fields from an earlier coarse pass over the raw corpus: the original raw category string, a short reason field, a keyword field, and a confidence score. These metadata are used only as classification inputs or normalization aids. Engagement outcomes such as reads, likes, shares, and *zaikan* never enter the classification prompt.

A.2 Taxonomy Construction and Discovery Review

Within the 2015–2024 archive, the original `category` field contains 862 distinct raw strings. Many already correspond to recognizable governance functions, but others are near-duplicates, wording variants, or generic placeholders. The final taxonomy fixes ten mutually exclusive detailed labels and four higher-order families. Table SI.2 reports the final taxonomy, the number of posts assigned to each label, the number of raw-label strings folded into each label, and each label's share of the cleaned corpus.

The codebook was frozen after a 400-title discovery review designed to surface both common labels and difficult boundary cases. As Table SI.3 shows, the first round balanced across five time

Table SI.2: Final ten-label taxonomy, higher-order families, and corpus shares.

detailed_label	family	n_posts	n_raw_labels	share_posts
Ideological Education	Hard Propaganda	470,939	49	0.112
Leadership and Political Events	Hard Propaganda	530,382	34	0.126
Public Service Information	Public Service	524,185	151	0.125
Social Protection and Welfare	Public Service	274,601	215	0.065
Emergency and Risk Communication	Public Service	544,084	44	0.129
Policy and Administrative Disclosure	State Governance	432,057	49	0.103
Social Governance and Enforcement	State Governance	203,340	90	0.048
Mass Mobilization and Participation	State Governance	96,840	40	0.023
Economic Development and Construction	Soft Propaganda	641,160	113	0.152
City Image and Cultural Activity	Soft Propaganda	489,484	81	0.116

bins and the ten cleaned labels, while the second round concentrated on generic raw labels, low-confidence titles, long-tail raw labels, and mixed-domain cases.

Table SI.3: Discovery-sample design used to refine category boundaries and codebook rules.

phase	target	rule
Round 1	200	Balanced grid across five time bins and the ten cleaned labels
Round 2A	50	Generic or low-information raw labels
Round 2B	50	Low-confidence coarse labels
Round 2C	50	Long-tail raw labels
Round 2D	50	Mixed-domain and boundary titles

That discovery review clarified the main inclusion and exclusion rules that later structure both human coding and model prompting. The hardest boundaries are not random. They recur where title wording mixes leadership reporting with development claims, routine service information with welfare substance, policy release with enforcement, emergency warning with ordinary governance control, and development performance with city branding. Table SI.4 summarizes the resulting coding cues and the corresponding title patterns.

Table SI.4: Illustrative coding cues and title patterns for the ten detailed labels.

detailed_label	family	decision_cue
Ideological Education	Hard Propaganda	Political learning, Party education, discipline, a
Leadership and Political Events	Hard Propaganda	Leaders, meetings, inspections, speeches, appo
Public Service Information	Public Service	Practical guidance, reminders, procedures, noti
Social Protection and Welfare	Public Service	Education, health, employment, subsidies, pens
Emergency and Risk Communication	Public Service	Warnings, hazards, disasters, epidemics, safety
Policy and Administrative Disclosure	State Governance	Policy texts, formal notices, administrative disc
Social Governance and Enforcement	State Governance	Enforcement, punishment, inspections, anti-fra
Mass Mobilization and Participation	State Governance	Calls for participation, volunteering, campaigns
Economic Development and Construction	Soft Propaganda	Projects, investment, industry, infrastructure, ag
City Image and Cultural Activity	Soft Propaganda	Tourism, culture, festivals, exhibitions, place br

Raw-label normalization then folded stable variants into the fixed taxonomy while leaving generic placeholders to be resolved at the row level. For example, 纪检监察与党风廉政 was normalized into ideological education, 公众服务信息 into public service information, and 人事任免 into leadership and political events. By contrast, low-information placeholders such as 其他, 新闻动态, and 未分类 were not trusted on their own and instead were resolved with the title, the short reason field, and the keyword field. The discovery review therefore served codebook construction and normalization, not supervised parameter training.

A.3 Human Coding Rules and Validation Sample

The validation audit drew 2,000 titles with sampling probabilities proportional to city-level corpus size, while retaining at least one title from every city in the corpus. This preserves exposure to the largest posting environments without collapsing the validation sample into only a handful of high-volume cities. Table SI.5 summarizes the resulting audit sample.

Table SI.5: Coverage summary for the 2,000-title validation sample.

metric	value
Annotated titles	2,000
Cities represented	329
Provincial-level units represented	31
Minimum titles per city	1
Median titles per city	5
Maximum titles per city	25

Two independent coders then assigned each title one and only one primary label and the corresponding higher-order family. The operative rule was to code the title's *main communicative purpose*, not the broadest noun phrase or the most politically salient word in isolation. In practice, this meant preferring public-service information over welfare only when the title mainly explained how to do something; preferring leadership and political events over policy disclosure when leaders, meetings, inspections, speeches, or appointments were the center of the title; separating policy disclosure from governance and enforcement by distinguishing rule release from rule enforcement; and separating development performance from city-image content by distinguishing projects and economic upgrading from tourism, festivals, and branding. Coders were blind to model output and to all engagement outcomes. Disagreements were adjudicated to produce the final gold-standard label. The 2,000-title audit sample was held out from prompt design and used only for validation.

A.4 Frozen LLM Prompt

The full-corpus pass used the OpenAI Responses API with `model = gpt-5.4` and `reasoning_effort = low` under a deterministic single-label instruction. The admissible Chinese primary labels were 意识形态与宣传教育, 时政与领导活动, 公共服务信息, 社会保障与公共福利, 应急管理与风险沟通, 政策与政务公开, 社会治理与执法通报, 群众动员与社会参与, 经济与发展建设, and 城市形象与文化活动. The production prompt was frozen before large-scale assignment and used for both the corpus pass and the validation audit. The prompt was:

Frozen prompt used for title-based coding

```
You are a Chinese political-communication annotator.
Task: classify a local-government WeChat post into a fixed ten-label taxonomy.
Use only title-side inputs. Do not use engagement metrics or infer category from popularity.
Choose exactly one primary label according to the post's main communicative purpose.
Decision rule: ask whether the title mainly teaches political values, reports leaders or political
events, provides practical service information, describes welfare or benefit provision, manages
immediate risk, discloses policy or administrative procedure, reports enforcement or governance action,
mobilizes participation, promotes development, or projects culture and city image.
Boundary rules:
- public service information vs. social protection and welfare: procedure and daily-use guidance vs.
welfare substance, health, education, jobs, subsidies, and pensions;
- leadership and political events vs. policy disclosure: leaders, meetings, inspections, speeches, and
appointments vs. policy texts, notices, budgets, and formal disclosure;
- policy disclosure vs. governance and enforcement: rule release vs. rule enforcement;
- emergency and risk communication vs. governance and enforcement: urgent warnings and hazards vs.
routine control and punishment;
- economic development vs. city image: projects and performance claims vs. tourism, festivals,
exhibitions, and branding.
Return JSON only in the following schema:
{"primary_label":"...", "family_label":"...", "confidence":0.00, "brief_rationale":"..."}
Input fields:
title = {title}
public_account_name = {public_account_name}
province = {province}
city = {city}
publish_date = {publish_date}
category_raw = {category_raw}
keywords = {keywords}
reason = {reason}
```

There was no fine-tuning stage and no supervised training split in the machine-learning sense. The discovery sample was used to freeze the codebook and prompt, after which the 2,000-title coder audit was held out for validation.

A.5 Validation Results

Table SI.6 reports the reliability of the dual-coded audit sample. Exact agreement reaches 0.8875 at the detailed-label level and 0.9130 at the family level. Cohen's κ is 0.8726 for the ten-label task and 0.8811 for the four-family task. Only 11.25% of titles required adjudication, which is consistent with the fact that the hardest disagreements cluster around a small set of expected boundary zones rather than around wholesale coder confusion.

Table SI.6: Inter-coder reliability for the dual-coded 2,000-title validation sample.

statistic	value
Primary-label exact agreement	0.8875
Family-label exact agreement	0.9130
Primary-label Cohen's kappa	0.8726
Family-label Cohen's kappa	0.8811
Adjudication rate	0.1125

Against the adjudicated gold-standard labels, the frozen LLM reaches 0.8150 accuracy and 0.7955 macro F1 on the ten-label task, as shown in Table SI.7. At the family level, accuracy rises to 0.8755 and macro F1 to 0.8642, reported in Table SI.8. This higher family-level performance is

substantively important because the paper's theory-facing analyses are conducted primarily at the family level.

Table SI.7: Overall validation metrics for detailed-label LLM coding.

statistic	value
Accuracy	0.8150
Macro F1	0.7955
Evaluation titles	2,000

Table SI.8: Overall validation metrics for family-level LLM coding.

statistic	value
Accuracy	0.8755
Macro F1	0.8642
Evaluation titles	2,000

Table SI.9 shows that family-level performance is strongest for soft propaganda and public service, while the weakest family is state governance, which is also the most semantically intermediate family in the codebook. Table SI.10 shows the same pattern at the ten-label level: performance is strongest for city-image and cultural content, economic development, leadership events, and emergency communication, and weaker for mass mobilization, policy disclosure, welfare content, and general public-service information. Those are also the labels where discovery review and coder adjudication revealed the greatest boundary pressure.

Table SI.9: Class-wise validation metrics for the four higher-order families.

family	precision	recall	f1	support
Hard Propaganda	0.8478	0.8966	0.8715	435
Public Service	0.8959	0.9016	0.8987	630
Soft Propaganda	0.9455	0.8691	0.9057	619
State Governance	0.7567	0.8070	0.7810	316

Table SI.10: Class-wise validation metrics for the ten detailed labels.

detailed_label	family	precision	recall	f1	support
Public Service Information	Public Service	0.6911	0.8019	0.7424	212
City Image and Cultural Activity	Soft Propaganda	0.9280	0.8974	0.9125	273
Emergency and Risk Communication	Public Service	0.8000	0.8848	0.8403	217
Ideological Education	Hard Propaganda	0.7304	0.9313	0.8187	160
Policy and Administrative Disclosure	State Governance	0.6875	0.7857	0.7333	168
Leadership and Political Events	Hard Propaganda	0.8750	0.8145	0.8437	275
Social Protection and Welfare	Public Service	0.8649	0.6368	0.7335	201
Social Governance and Enforcement	State Governance	0.8247	0.8081	0.8163	99
Economic Development and Construction	Soft Propaganda	0.9115	0.8035	0.8541	346
Mass Mobilization and Participation	State Governance	0.6667	0.6531	0.6598	49

Figure SI.1 visualizes the family-level confusion matrix. Most remaining error is not a collapse of propaganda and service into one undifferentiated category. Instead, it is concentrated in the expected intermediate zone where state-governance titles can be absorbed into adjacent categories. That pattern is consistent with the codebook logic and helps justify using the four families for diagnostics while reserving the three main families for the paper's core theoretical tests.

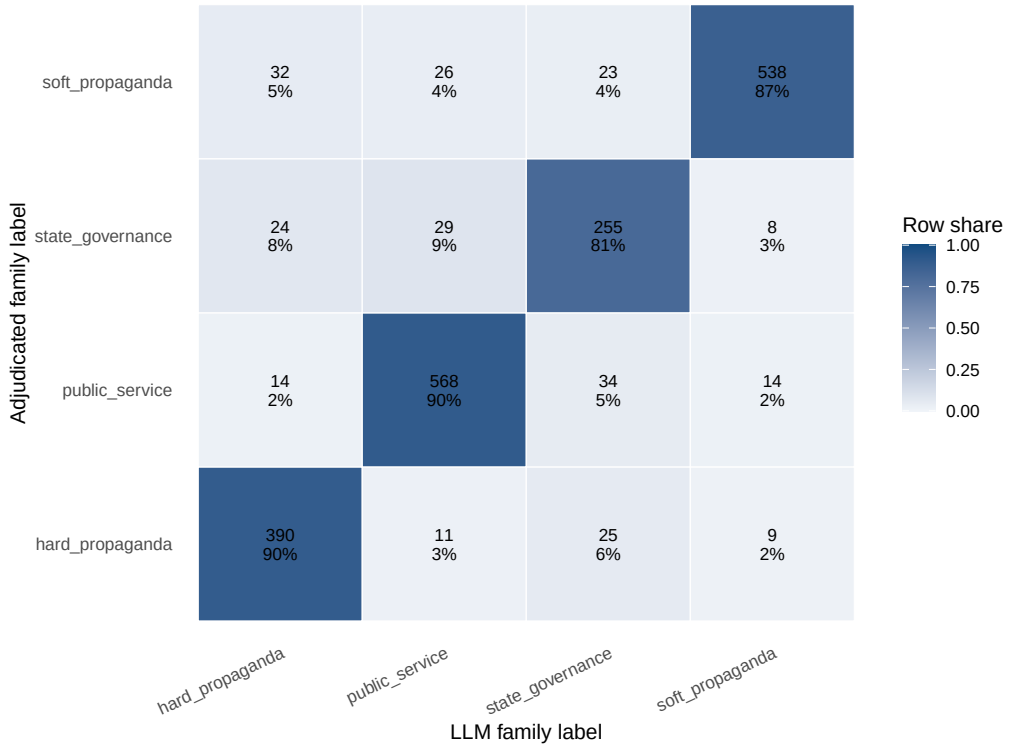

Figure SI.1: Family-Level Confusion Matrix for the WeChat Classification Audit